



CeylonRail Intelligence

AI-Driven Intelligent Railway Operations System for Sri Lanka

Project ID: R26-IT-127

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Team Members

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Problem Statement





Users and Stakeholders

Primary Users:

- Train passengers
- Ticket inspectors
- Railway operators

Stakeholders:

- Sri Lanka Railways
- Transport policymakers
- Government authorities



Impact of the Problem

Passenger Impact:

- Long waiting times
- Uncertain arrival times

Operational Impact:

- Inefficient resource utilization
- High inspection workload

Economic Impact:

- Revenue loss from fraud

Existing Systems

- Static train scheduling
- Manual ticket inspections
- Basic passenger information systems
- Rule-based seat allocation
- Traditional forecasting models (ARIMA)

Limitations

- Lack of AI-driven decision support
- No real-time delay prediction
- Weak fraud detection mechanisms
- Poor passenger demand forecasting
- Inefficient train scheduling

Existing Systems and Limitations



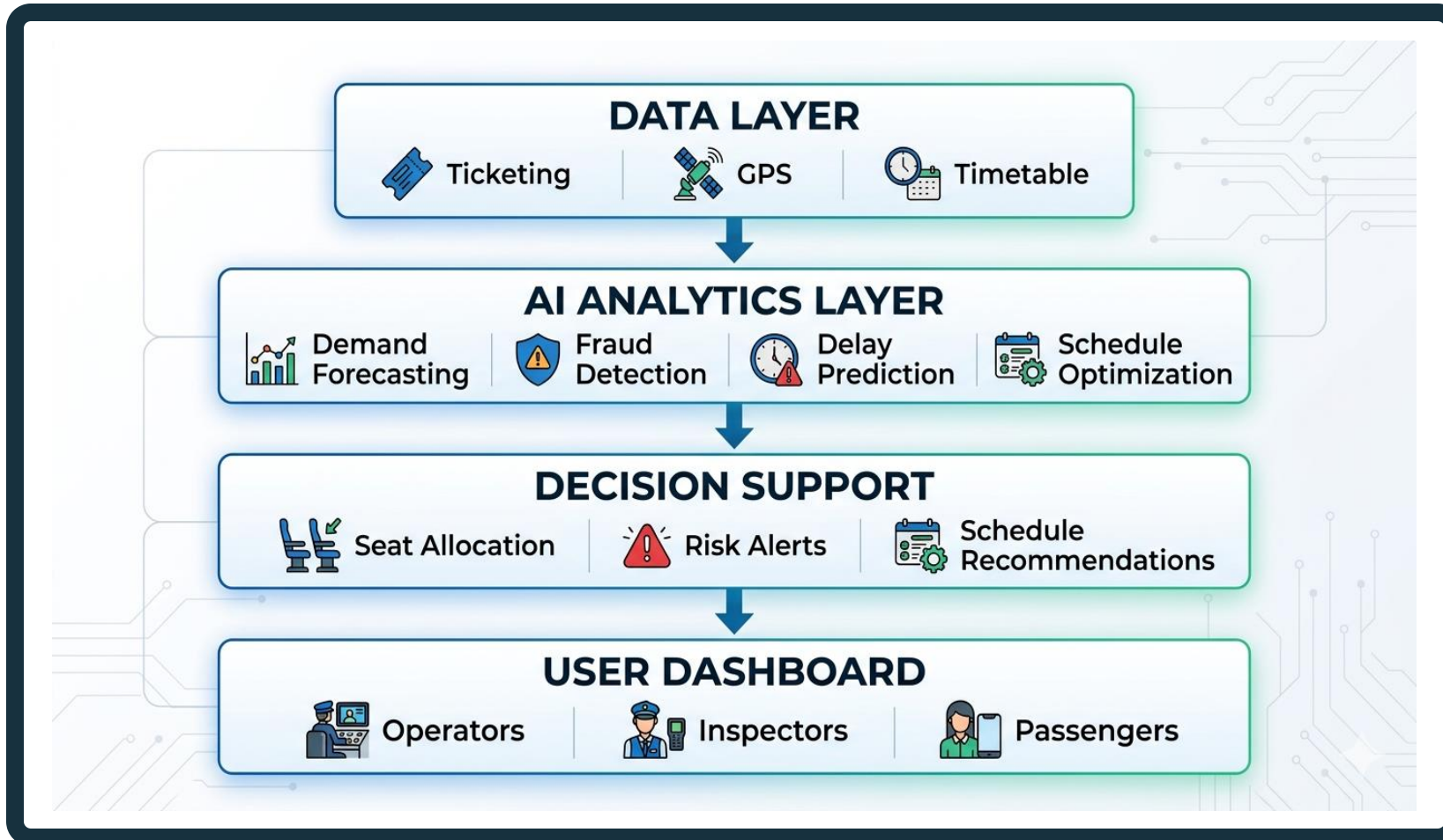
Proposed Solution



An AI-driven intelligent railway decision support system integrating:

- Passenger demand forecasting
- Adaptive seat allocation
- Fraud detection
- Train delay prediction
- Train scheduling optimization

System Overview



Individual Components



Figure 1: System Overview Diagram

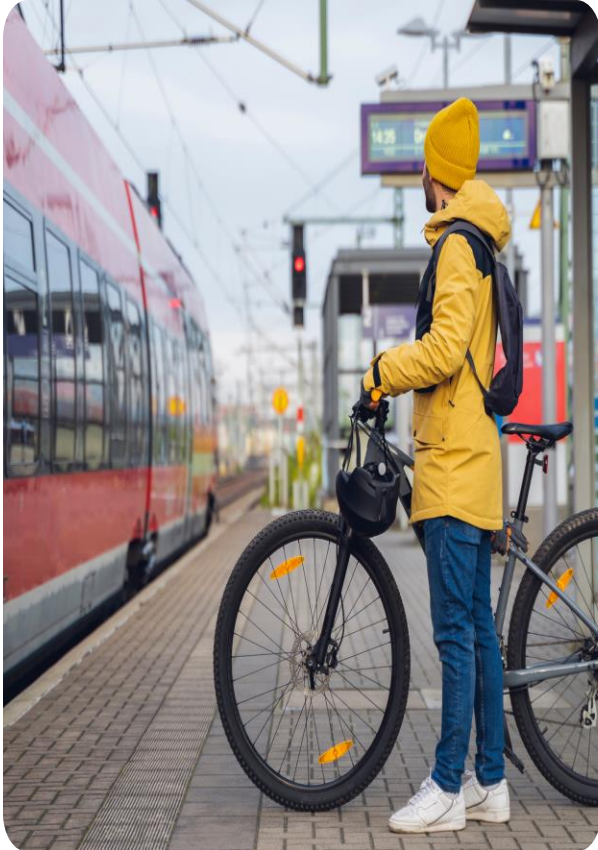
1. Multi-Objective Train Scheduling Optimization
2. Real-Time Train Tracking and Delay Prediction
3. Intelligent Ticket Fraud Detection
4. Passenger Demand Forecasting & Seat Allocation

Multi-Objective Train Scheduling Optimization

IT22217240 | Narasinghe N.K.B.N.N



Multi-Objective Train Scheduling Optimization



Optimizes train schedules to improve efficiency, reduce congestion, and balance passenger demand.

Novelty: Uses evolutionary algorithms (NSGA-II) to solve multiple conflicting objectives, unlike traditional fixed schedules.

Problem Definition:

- Static schedules cannot adapt to dynamic passenger flows and operational constraints.



Multi-Objective Train Scheduling Optimization

Objective: Develop an optimization framework for adaptive train scheduling.

Sub-Objectives:

- Model constraints: train capacity, track availability, maintenance schedules.
- Generate schedules optimizing delay, resource usage, and passenger comfort.
- Evaluate performance under different demand scenarios.



Multi-Objective Train Scheduling Optimization

Research Methodology:

- Operations research and optimization in transportation systems.

Technologies Used:

- NSGA-II (Non-dominated Sorting Genetic Algorithm).
- Simulation for scenario testing.



Multi-Objective Train Scheduling Optimization

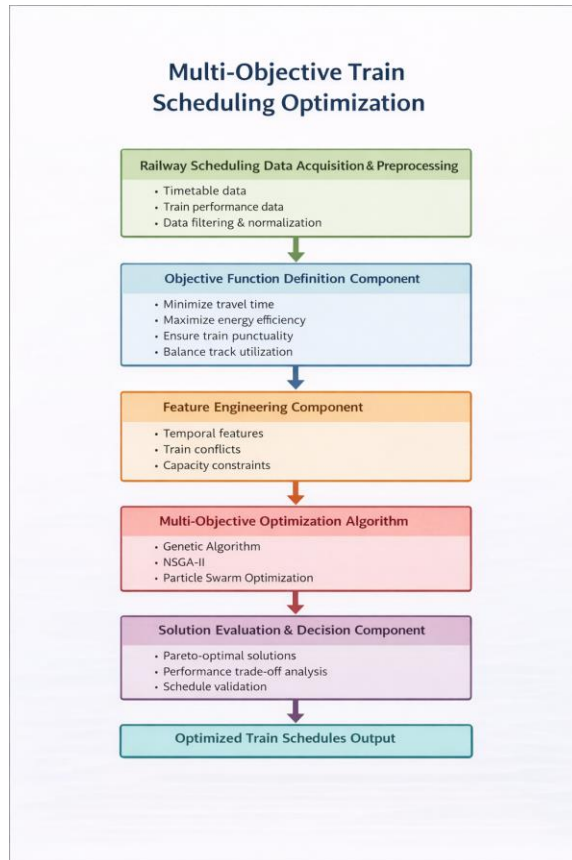
Validation & Evaluation:

- Compare optimized schedules vs existing schedules.
- Metrics: On-time performance, train utilization, passenger wait times.

Data & Ethics:

- Timetable, train capacity, and route constraints used.
- No personal data required; ethical clearance for operational data usage.

Multi-Objective Train Scheduling Optimization



System Overview:

- Optimization engine integrated with demand forecasting and delay prediction modules.

Dependencies:

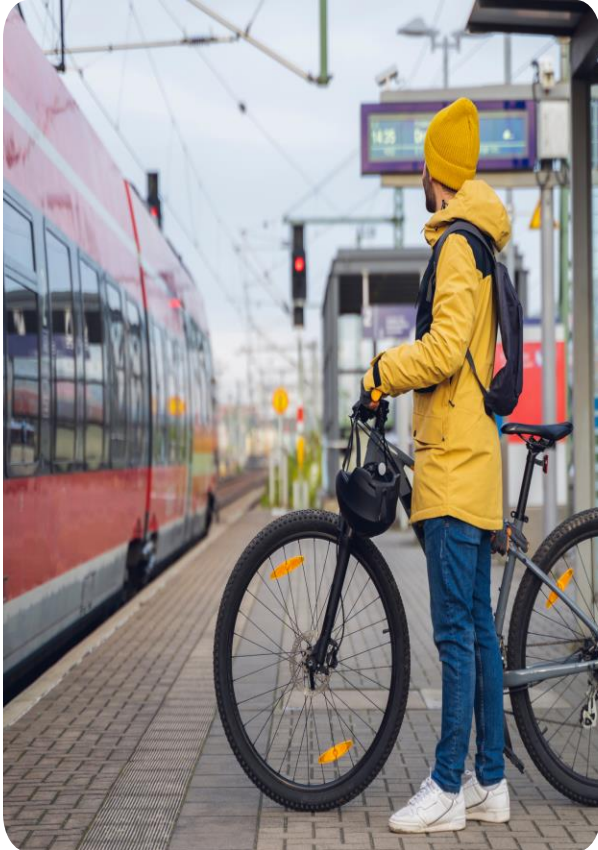
- Relies on outputs from passenger demand forecasting and real-time delay modules.

Real-Time Train Tracking & Delay Prediction

IT22051202 | Didulantha C.S



Real-Time Train Tracking & Delay Prediction



Provides accurate arrival time predictions to reduce passenger waiting and operational inefficiencies.

Novelty: Uses probabilistic ML models to predict delays and quantify uncertainty, unlike traditional static timetable approaches.

Problem Definition:

- Existing systems rely on scheduled arrival times only.
- Frequent unpredicted delays cause passenger dissatisfaction and resource mismanagement.



Real-Time Train Tracking & Delay Prediction

Objective: Predict train arrival times accurately and in real-time.

Sub-Objectives:

- Collect and preprocess train GPS and timetable data. Develop ML models for delay prediction. Generate real-time alerts for operators and passengers.



Real-Time Train Tracking & Delay Prediction

Research Methodology:

- Time-series analysis and predictive modeling in railway operations.

Technologies Used:

- LSTM, GRU, regression-based probabilistic models.



Real-Time Train Tracking & Delay Prediction

Validation & Evaluation:

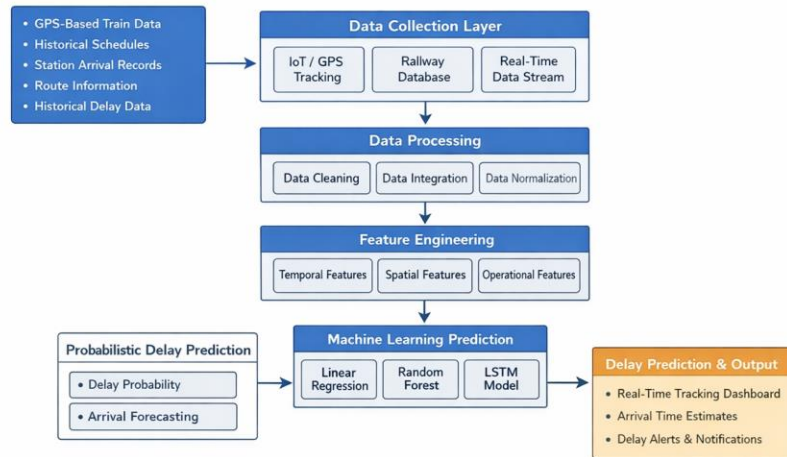
- Compare predicted vs actual arrival times.
- Metrics: Mean Absolute Error (MAE), RMSE, Prediction Interval Coverage.

Data & Ethics:

- GPS and train logs from Sri Lanka Railways; anonymized where needed.

Real-Time Train Tracking & Delay Prediction

Intelligent Real-Time Train Tracking and Probabilistic Delay Prediction Framework



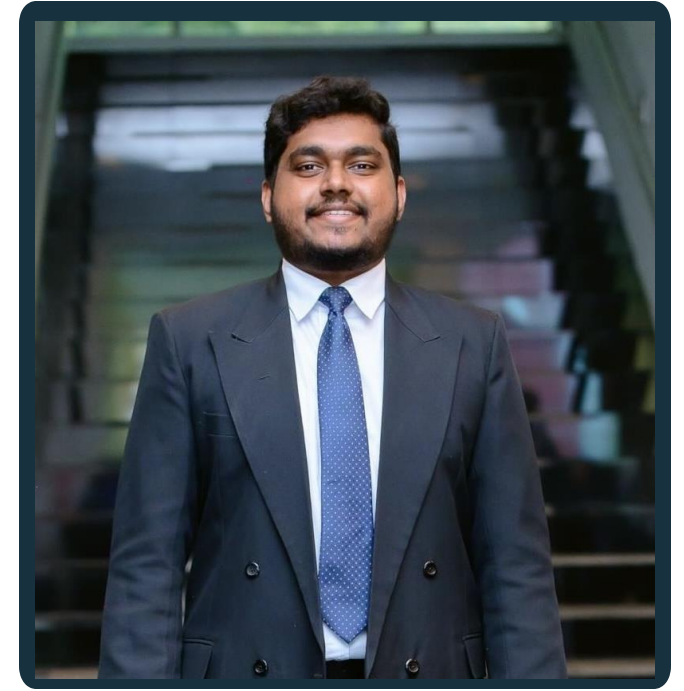
System Overview:

- Real-time data ingestion from GPS sensors.
- Predictive model outputs feed into the decision support dashboard.

Dependencies:

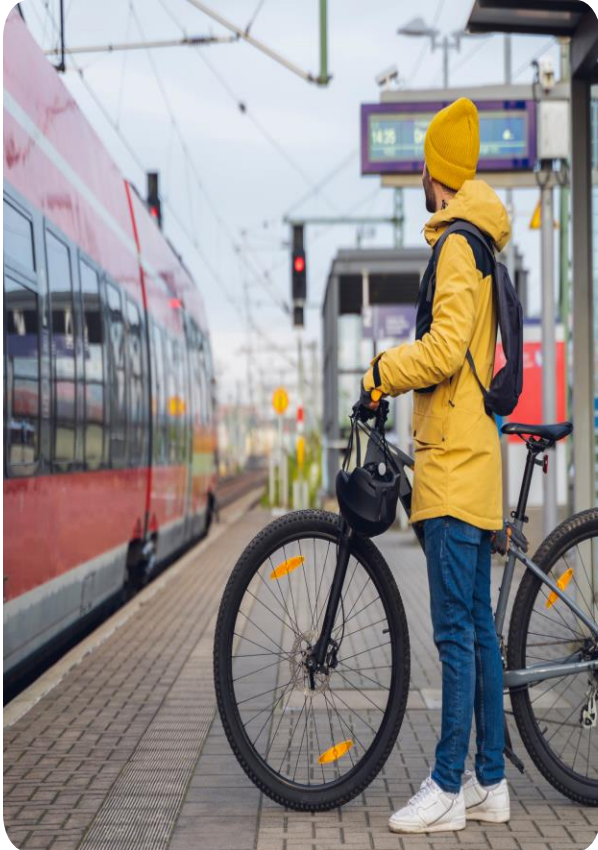
- Requires access to train locations and timetable data.
- Integrated with scheduling and seat allocation modules.

Advanced Intelligent Ticket Fraud Detection and Risk-Aware Passenger Verification



IT22126610 – Muthukudaarachchi V.U

Advanced Intelligent Ticket Fraud Detection and Risk-Aware Passenger Verification



Detects fraudulent ticket usage to reduce revenue loss and improve fairness in railway operations. Supports proactive inspection and decision-making.

Novelty: Combines anomaly detection with supervised machine learning to detect complex fraud patterns that are undetectable by traditional methods.

Problem Definition:

- Manual ticket inspection is inefficient and cannot detect sophisticated fraud.
- Revenue loss due to concession misuse and duplicate tickets.



Advanced Intelligent Ticket Fraud Detection and Risk- Aware Passenger Verification

- Research Objective & Sub-Objectives: Objective: Develop an AI-based system to detect suspicious ticket activities in real-time.
- Sub-Objectives: Analyze historical ticket transactions to identify patterns. Apply anomaly detection and classification techniques to flag fraud. Provide actionable alerts to railway operators.



Advanced Intelligent Ticket Fraud Detection and Risk-Aware Passenger Verification

Research Methodology:

- Data mining and machine learning in the transportation domain.
- Knowledge of ticketing operations and railway policies applied to model fraud behavior.

Technologies Used:

- Isolation Forest, Autoencoders for anomaly detection.
- Random Forest and XGBoost for classification.



Advanced Intelligent Ticket Fraud Detection and Risk-Aware Passenger Verification

Validation & Evaluation:

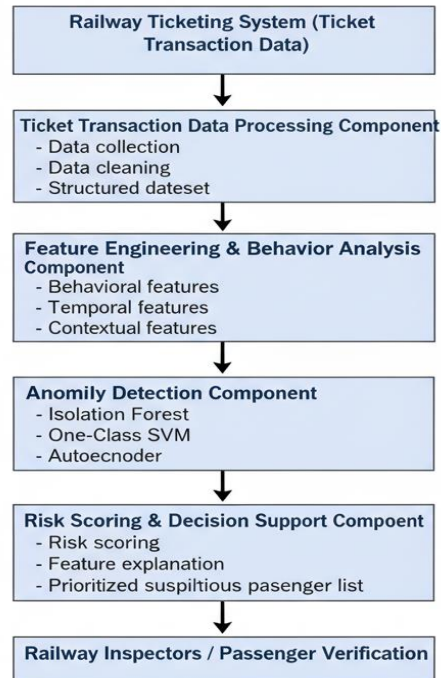
- Compare flagged tickets against ground truth from historical fraud cases.
- Metrics: Accuracy, Precision, Recall, F1-Score.

Data Availability & Ethics:

- Historical ticket logs available from Sri Lanka Railways.
- User privacy maintained; anonymized datasets used.

Advanced Intelligent Ticket Fraud Detection and Risk-Aware Passenger Verification

Advanced Intelligent Ticket Fraud Detection and Risk-Aware Verification System Architecture



System Overview:

- Component integrates with ticketing database.
- Processes data in real-time to detect anomalies.

Dependencies:

- Relies on accurate ticket transaction logs.
- Communicates with delay prediction and scheduling modules for operational context.

Advanced Spatiotemporal Demand Forecasting and Adaptive Seat Allocation using AI



IT22151506 – D.M.N.T Senawirathna

Advanced Spatiotemporal Demand Forecasting and Adaptive Seat Allocation using AI



Predicts passenger demand and dynamically allocates seats to reduce overcrowding and improve comfort.

Novelty: Combines spatio-temporal deep learning with optimization for real-time adaptive seat allocation.

Problem Definition:

- Manual or static seat allocation leads to overcrowding and passenger dissatisfaction.



Advanced Spatiotemporal Demand Forecasting and Adaptive Seat Allocation using AI

Objective: Forecast passenger flows and allocate seats dynamically.

Sub-Objectives:

- Use historical ticket data and seasonal trends for demand forecasting.
- Allocate seats based on predicted passenger flow.
- Adapt allocations in real-time during peak periods.



Advanced Spatiotemporal Demand Forecasting and Adaptive Seat Allocation using AI

Research Methodology:

- Time-series forecasting and optimization in railway operations.

Technologies Used:

- LSTM, GRU, Temporal CNN for prediction. Linear programming or heuristic algorithms for seat allocation.



Advanced Spatiotemporal Demand Forecasting and Adaptive Seat Allocation using AI

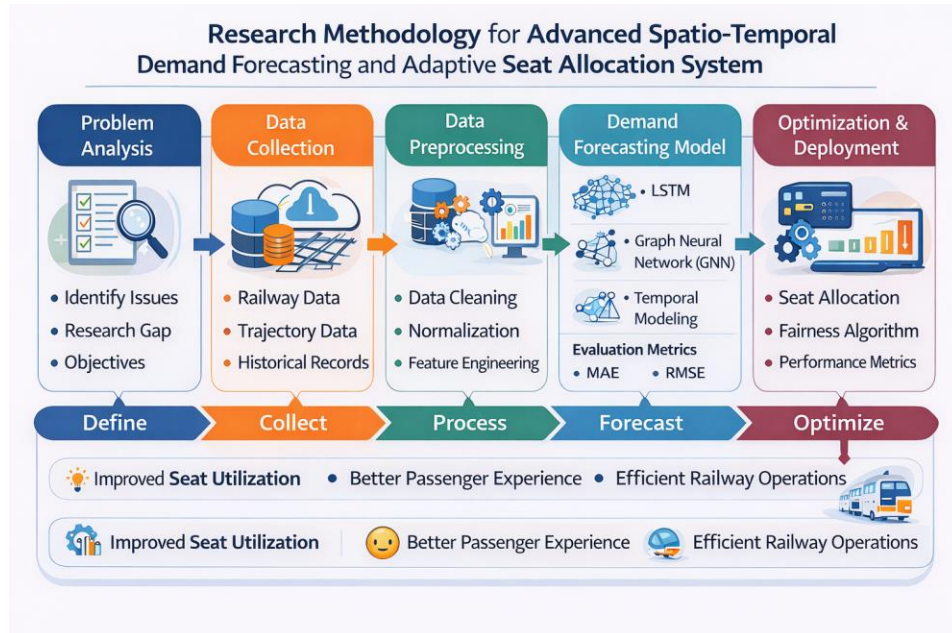
Validation & Evaluation:

- Forecast accuracy: MAE, RMSE.
- Seat allocation efficiency: occupancy rate, passenger satisfaction surveys.

Data & Ethics:

- Historical ticketing data, anonymized for privacy. Ensures compliance with ethical standards for passenger data.

Advanced Spatiotemporal Demand Forecasting and Adaptive Seat Allocation using AI



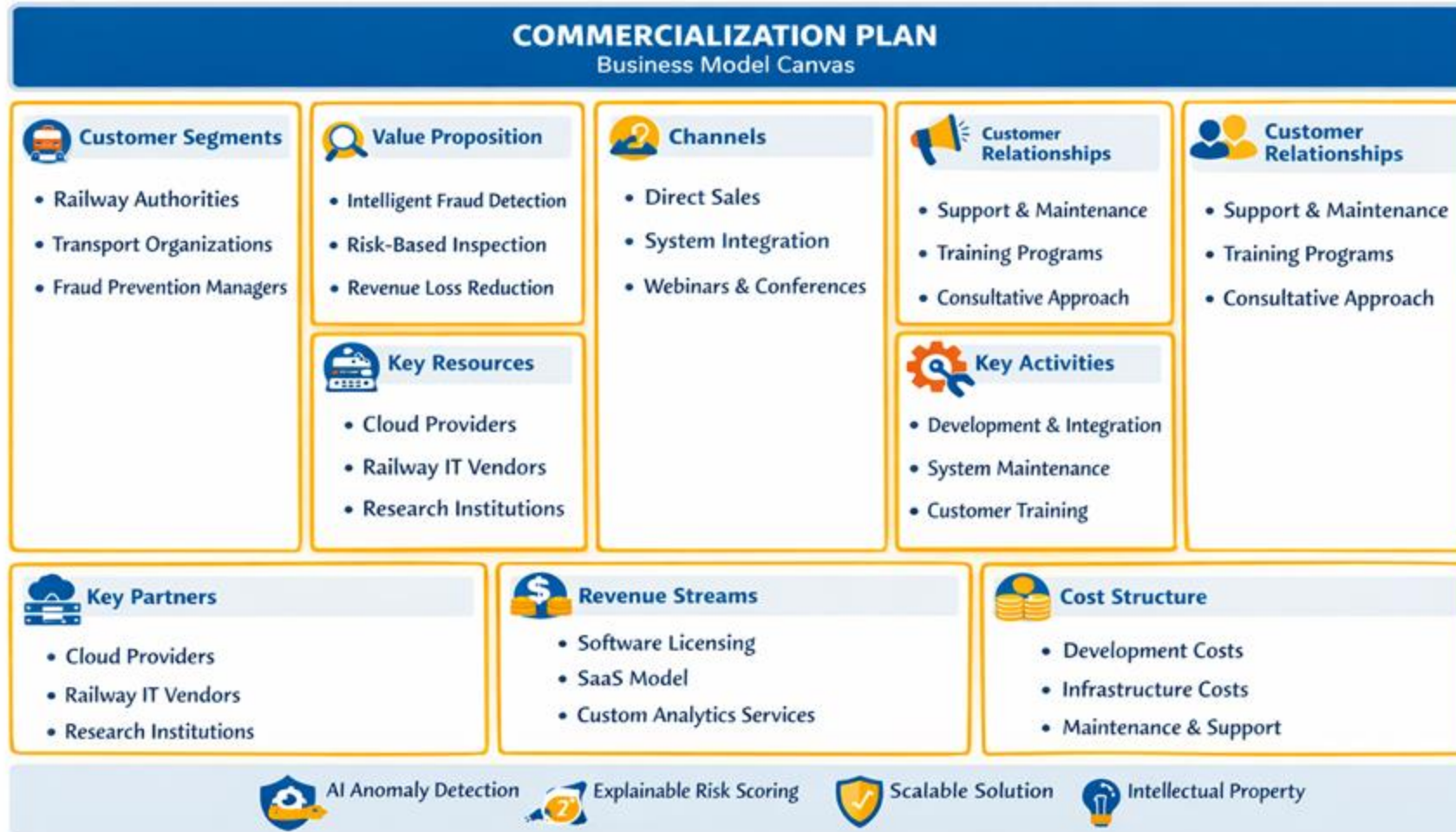
System Overview:

- Component works alongside ticketing and scheduling systems.
- Feeds predicted demand into scheduling and seat allocation modules.

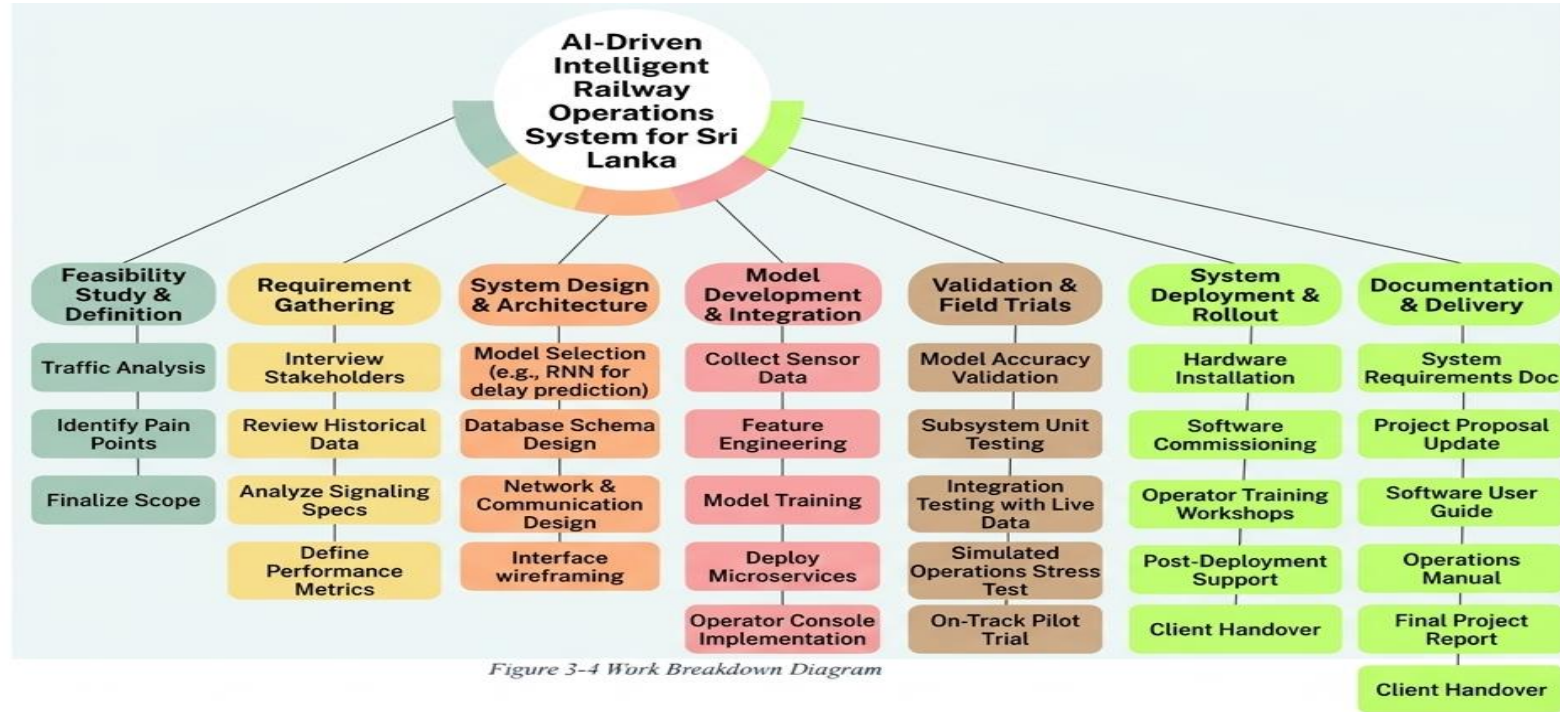
Dependencies:

- Requires ticket transaction history and timetable info.
- Interacts with fraud detection and scheduling modules for integrated decision-making.

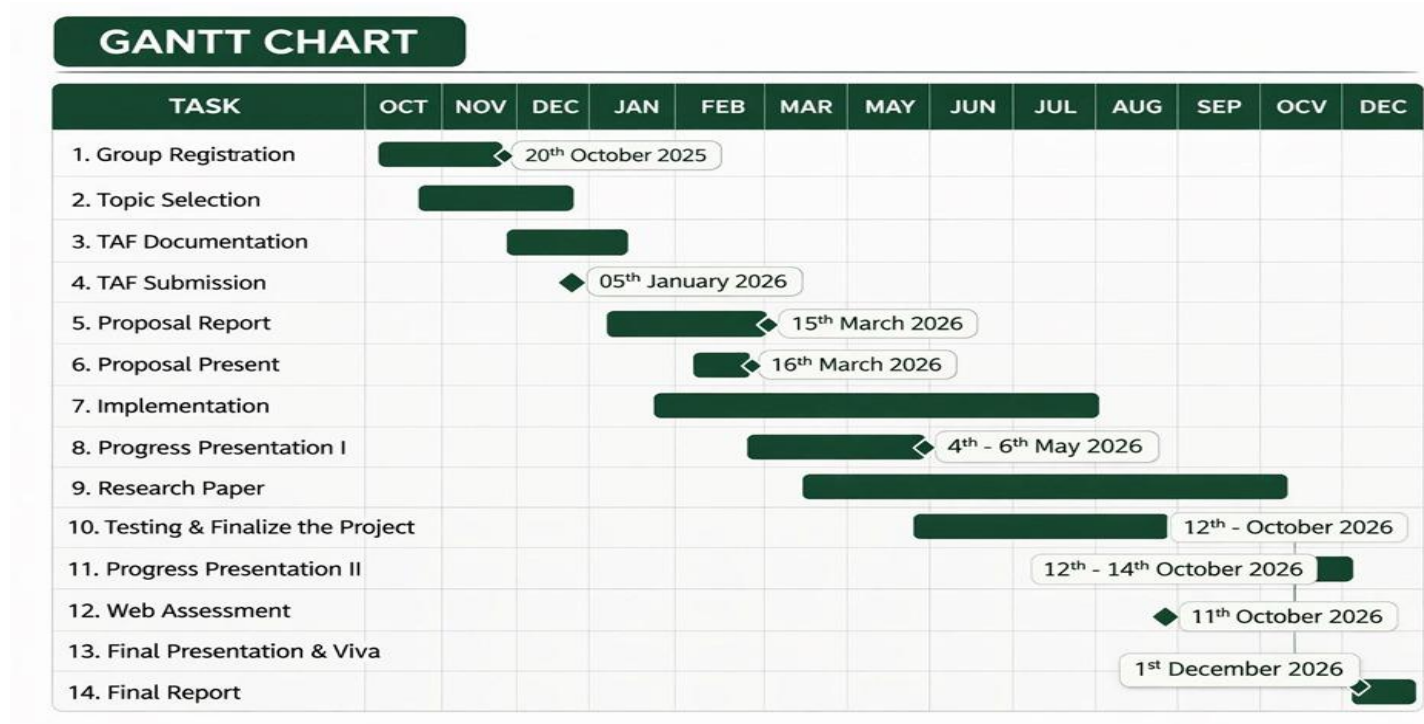
Commercialization Potential



Work Breakdown Structure



Project Timeline





Thankyou!